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Rank-Order Assignment Project Plan

Part I: Assumption and Mathematical Formulation

Problem and Assumptions: The goal is to assign N doctors to K hospitals based on the doctors' ranked preferences. Each doctor provides a ranked list of hospitals, with rank 1 representing their most preferred hospital. Each hospital has a fixed maximum capacity, and the total capacity across all hospitals is assumed to be sufficient to assign every doctor. Each doctor must be assigned to exactly one hospital, hospitals do not express preferences for doctors, and hospitals do not need to be filled to capacity.

Mathematical Formulation: The doctor preference rankings are represented by r_{nk} for integers $n \leq N$ and $k \leq K$, which represents doctor n's ranking of hospital k. The value ranges from 1 to K, with 1 being the most preferred hospital. The maximum capacity of each hospital is represented by c_k. The unknown assignment variable x_{nk} equals 1 if doctor n is assigned to hospital k, and 0 otherwise. The objective is to minimize the sum of the preference ranks associated with all assignments. Each doctor must be assigned to exactly one hospital, and the total number of doctors assigned to a hospital cannot exceed its capacity.

Part II: Selected Algorithm and Baseline

Selected algorithm: The Hungarian method will be used to find the assignment that minimizes the total preference rank across all doctors (Kuhn, 1955). Doctors' hospital rankings will serve as assignment costs, with lower ranks representing preferred assignments. Because the traditional Hungarian method uses one-to-one assignments, hospitals with capacities greater than one will be represented as multiple identical slots. For example, a hospital with a capacity of four will have four slots with the same preference cost for a given doctor. If total hospital capacity exceeds the number of doctors, dummy doctors with zero preference costs will be added as needed to create a square cost matrix. The resulting doctor-to-slot assignments will then be converted back to doctor-to-hospital assignments.

Chosen baseline: A greedy assignment approach will serve as the baseline. Doctors will be considered sequentially and assigned to their highest-ranked hospital with remaining capacity (Garcia, 2025). Unlike the Hungarian method, the greedy approach makes locally optimal decisions without considering the overall assignment. The methods will be compared using total preference rank, average assigned rank, and percentage of doctors receiving their first choice.

Part III: References

García, A. Greedy algorithms: a review and open problems. J Inequal Appl 2025, 11 (2025). https://doi.org/10.1186/s13660-025-03254-1

Kuhn, H.W. (1955), The Hungarian method for the assignment problem. Naval Research Logistics, 2: 83-97. https://doi.org/10.1002/nav.3800020109

Part IV: Limitations and Potential Extensions

Possible Limitations: The Hungarian algorithm requires hospitals with larger capacities to be represented as multiple slots. Rankings capture the order but not strength of preferences, and minimizing total rank may not ensure fairness for every doctor. The model also assumes complete rankings and considers only doctors' preferences.

Extensions: Extensions could use ratings instead of rankings to capture preference strength, introduce a fairness-based objective that penalizes poor assignments more heavily, allow incomplete preference lists, or compare the Hungarian method with auction or transportation algorithms.